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Department of Computer Science

Final Report

RAG Chatbot — MILOUDI — NLP

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1. What is RAG?

Retrieval-Augmented Generation (RAG) combines document retrieval with language model generation. Instead of inventing answers, the LLM first retrieves relevant passages from your dataset, then generates a response based only on that context.

Pipeline:

- Load dataset (`data/comments.txt`)
- Split into chunks (800 chars, overlap 80)
- Generate embeddings (`all-MiniLM-L6-v2`, 384D vectors)
- Store in FAISS (local vector database)
- User question → embedding → FAISS search (Top-3 chunks)
- Prompt LLM with retrieved context
- Generate answer

2. Construction Steps

1. Create dataset of customer comments.
2. Load with LangChain TextLoader.
3. Split into chunks (size 800, overlap 80).
4. Generate embeddings with `sentence-transformers/all-MiniLM-L6-v2`.
5. Store vectors in FAISS for cosine similarity search.
6. Query LLM via HuggingFace Inference API.
7. Deploy interface with Gradio.

3. LLM Used

Model: `WORKING_MODEL` (fallback: RAG retrieval, no LLM API). Architecture: Transformer decoder. Free via HuggingFace Inference API. Migrated from Flan-T5 to Mistral-7B-Instruct in 2025.

4. Embedding Model

Name: `sentence-transformers/all-MiniLM-L6-v2`. Architecture: Distilled BERT (MiniLM). Output: 384-dimensional normalized vectors. Runs locally in Colab.

5. Vector Database

Tool: FAISS (Facebook AI Similarity Search). Search: Cosine similarity, Top-3 chunks.
Advantages: Fast, free, offline.

6. Dataset

File: `data/comments.txt`. Content: Customer e-commerce feedback (products, delivery, service, statistics).

7. Problems Encountered

- **P1:** Numpy conflict in Colab Python 3.12. *Solution:* Reinstall `numpy==1.26.4`.
- **P2:** Flan-T5 removed from HuggingFace free API. *Solution:* Switch to Mistral-7B-Instruct.
- **P3:** HuggingFace token missing API permission. *Solution:* Create new READ token with API access.

8. GitHub Submission

Repository: Melouka-mld/RAG-Chatbot-NLP_MILOUDI. Files:

- Notebook: `MILOUDI_RAG_CHATBOT_VERIFIED.ipynb`
- Scripts: `ingestion.py`, `retrieval.py`, `chatbot_rag.py`
- Dataset: `data/comments.txt`
- `requirements.txt`, `README.md`

9. Evaluation Results

Question	Answer (Summary)
Delivery speed	Customers often complain about delayed shipping, wrong/broken items, poor service response.
Common complaints	Same issues: delays, wrong items, poor service.
Product quality	Dissatisfaction due to broken/wrong items.
Customer service	Negative: no replies to emails, phone line always voicemail.
Refund process	Positive: issue resolved in under 10 minutes.

10. Screenshots

AVG TIME: 0.05

	Question	Answer	Method	Time(s)	Chunks	
0	What do customers say about delivery speed?	Most Common Complaints:\n- Delayed shipping beyond promised dates\n- Wrong o...	RAG retrieval	0.0	3	
1	What are the most common complaints?	Most Common Complaints:\n- Delayed shipping beyond promised dates\n- Wrong o...	RAG retrieval	0.0	3	
2	How satisfied are customers with product quality?	Most Common Complaints:\n- Delayed shipping beyond promised dates\n- Wrong o...	RAG retrieval	0.0	3	
3	How was the customer service described?	Service Comment 4 (Negative):\nSent 4 emails over 2 weeks with no reply what...	RAG retrieval	0.0	3	
4	How long did the refund process take?	=== CUSTOMER SERVICE FEEDBACK ===\n\nService Comment 1 (Positive):\nThe cust...	RAG retrieval	0.0	3	

saved: evaluation_results.csv

Figure 1: Evaluation table generated by the RAG pipeline.

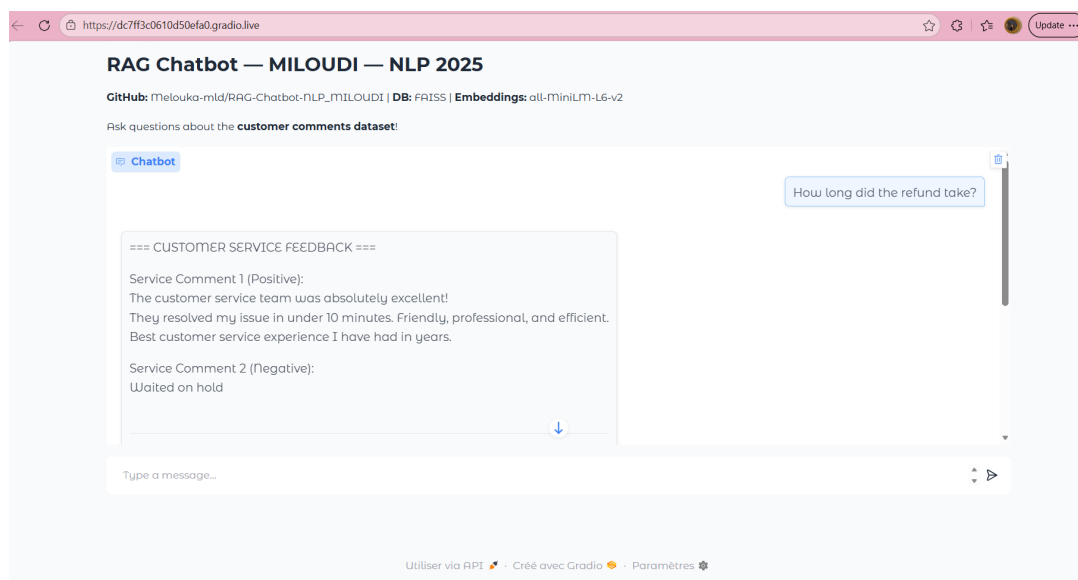


Figure 2: Chatbot interface showing customer feedback Q&A.